Capturing Fine-Scale Turbulence in Planet-Forming Disks Using CNN-Augmented Fluid Solvers

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Abstract

Understanding turbulence is crucial for studying gas-dust dynamics in planet-forming disks, thereby addressing planet formation. However, fully resolving these small-scale effects often remains beyond the capabilities of current numerical facilities. To address this, we developed a hybrid model based on Kochkov et al. (2021), combining traditional numerical methods for fluids dynamics with deep learning to capture fine-scale dynamics. Specifically, we apply a convolutional neural network (CNN) after each solver step to correct for discretisation errors. The CNN is trained using a solver-in-the-loop training approach, which integrates a differentiable fluid solver into the optimization of network parameters. We first tested the model on 2D homogeneous isotropic turbulence and found that the accuracy of the neural network improves when exposed to longer simulation timesteps during training. This is consistent with previous findings. We then aim to extend this approach to shearing box simulations, ultimately enabling the study of local turbulence in planet-forming disks.

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